src/dynamics.py · energy MMA3001 DeepONet project

energy

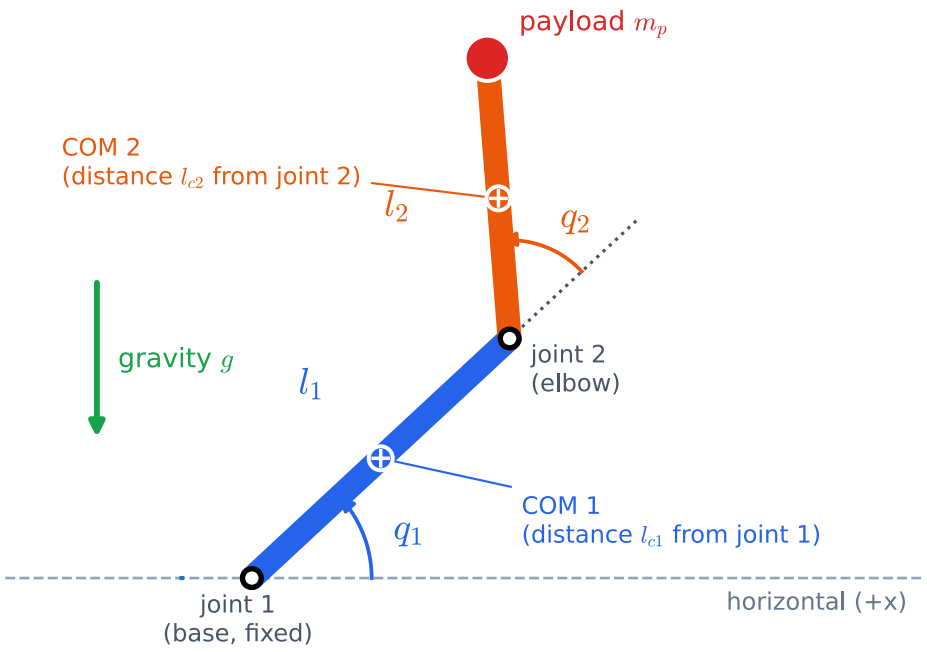
Kinetic + potential energy, the power balance dE/dt = τ·q̇ − b|q̇|², and how it checks the whole model

In this document

1. The question
2. Toolbox: the two energies
3. The power-balance law
4. Energy as a debugging tool
5. Checking with friction and a motor
6. Worked example
7. Code-maths dictionary

The set-up: a two-link robot arm in a vertical plane

Everything in this project describes one physical object: a planar robot arm with two rigid links, two motorised joints, bolted to a wall at the origin. It moves in a *vertical* plane, so gravity matters. At the tip of link 2 we may clip on an extra weight (the **payload**). Each joint has a little friction that slows it down in proportion to how fast it spins.



The arm. Blue = link 1, orange = link 2. The angle q_1 is measured from the horizontal; the angle q_2 is measured from the *straight-on extension* of link 1 (dotted), not from the horizontal. The dots with a white cross are the links' centres of mass (COM).

Conventions (fixed everywhere in the project): the state is $x = [q_1, q_2, \dot{q}_1, \dot{q}_2]$; gravity points in the $-y$ direction; potential energy is zero at the height of joint 1; positive angles and torques are anticlockwise.

Symbol	Meaning	Value in the code	Unit
$l_1,\ l_2$	link lengths (joint to joint)	0.5, 0.4	m

$m_1,\ m_2$	link masses (without payload)	2.0, 1.5	kg
$l_{c1},\ l_{c2}$	distance from a link's joint to its own centre of mass (COM)	0.25, 0.2 (middle of each link)	m
$I_1,\ I_2$	moment of inertia of a link about its own COM	$ml^2/12$ (uniform rod)	kg m^2
g	gravitational acceleration	9.81	m/s^2
m_p	point-mass payload at the tip of link 2 (varies per sample)	0 to 1 (dataset), up to 1.5 out-of-range	kg
b	viscous joint friction coefficient (varies per sample)	0.05 to 0.5	N m s/rad
$q_1,\ q_2$	joint angles: q_1 from the horizontal, q_2 relative to link 1	state	rad
$\dot{q}_1,\ \dot{q}_2$	joint angular velocities (a dot means d/dt)	state	rad/s
$\tau_1,\ \tau_2$	motor torques applied at joints 1 and 2 (the inputs)	$A\sin(2\pi ft+\phi)$	N m

Toolbox: Lagrangian mechanics in five minutes

Quick recap (the complete explanation is in the *mass_matrix* document). Write the **kinetic energy** T (energy of motion) and the **potential energy** V (energy stored in height). The Euler-Lagrange equation for each joint angle q_k is

$$\frac{d}{dt}\left(\frac{\partial T}{\partial \dot{q}_k}\right) - \frac{\partial T}{\partial q_k} + \frac{\partial V}{\partial q_k} = Q_k,$$

where Q_k is the outside "push" on joint k (motor torque, friction). It is just Newton's second law in disguise, but it never makes us think about the forces inside the joints.

1 The question this function answers

Energy is one single number that summarises the "motion budget" of the arm: how much is stored in speed (kinetic) and how much in height (potential). It matters here for two reasons. (i) It is a **physics-based correctness check**: with no friction and no motor, a correct model must conserve it, so any drift exposes a bug in the dynamics formulas or in the solver. (ii) It is a convenient quantity to plot in the visualiser (`viz.py` imports `energy`) and to judge neural-network predictions with.

```
def energy(x, p):
    q = x[:2];  qdot = x[2:]
    q1, q2 = q
    m2e, lc2e, _ = combine_payload(p.m2, p.lc2, p.I2, p.l2, p.m_p)
    T = 0.5 * qdot @ mass_matrix(q, p) @ qdot
    V = p.g * ( p.m1*p.lc1*np.sin(q1) + m2e*(p.l1*np.sin(q1) + lc2e*np.sin(q1 + q2)) )
    return float(T + V)
```

2 Toolbox: the two kinds of mechanical energy

- **Kinetic energy** T: energy of motion. For a point mass $\frac{1}{2}mv^2$; for something spinning $\frac{1}{2}I\omega^2$ (derived in *combine_payload*). For our arm both effects are present in every link, and when everything is added up (the long derivation in *mass_matrix*) it collapses to the compact form

  $$T = \tfrac{1}{2}\,\dot{q}^\top M(q)\,\dot{q} = \tfrac{1}{2}M_{11}\dot{q}_1^2 + M_{12}\dot{q}_1\dot{q}_2 + \tfrac{1}{2}M_{22}\dot{q}_2^2,$$

  which is exactly the line `0.5 * qdot @ mass_matrix(q, p) @ qdot` (`@` is matrix multiplication). It is never negative because M is positive definite.
- **Potential energy** V: energy stored in height, mgy. We derived it in *gravity_vector*: $V = g\big[m_1 l_{c1}\sin q_1 + m_{2e}(l_1\sin q_1 + l_{c2e}\sin(q_1+q_2))\big]$, the second code line. Heights are measured from joint 1, so $V < 0$ when the arm hangs below the base (and that is fine: only *changes* in V matter).

The total is $E = T + V$. Payload: the same m_{2e}, l_{c2e} from `combine_payload` appear in both parts (T through M, V explicitly). The inertia I_{2e} is needed only inside M, which is why the code throws it away with `_` in the unpacking.

3 The law that makes E useful: the power balance

How does E change in time? Differentiate $E = \frac{1}{2}\dot{q}^\top M\dot{q} + V(q)$ (product and chain rules):

$$\frac{dE}{dt} = \dot{q}^\top M\ddot{q} + \tfrac{1}{2}\dot{q}^\top\dot{M}\dot{q} + \dot{q}^\top G.$$

(Here $\partial V/\partial q = G$, and the two M terms come from differentiating $M\dot{q}\cdot\dot{q}$: $\ddot{q}$ once, and M itself changing once.) Now substitute the equation of motion $M\ddot{q} = \tau - c - G - b\dot{q}$:

$$\frac{dE}{dt} = \dot{q}^\top(\tau - c - G - b\dot{q}) + \tfrac{1}{2}\dot{q}^\top\dot{M}\dot{q} + \dot{q}^\top G = \dot{q}^\top\tau - b\,\dot{q}^\top\dot{q} + \Big[\tfrac{1}{2}\dot{q}^\top\dot{M}\dot{q} - \dot{q}^\top c\Big].$$

The gravity terms cancelled. The bracket is the key identity of mechanics: **it is zero**. We can check it with the explicit formulas from the *coriolis_vector* document. With $\dot{M}_{11} = -2h\dot{q}_2$, $\dot{M}_{12} = -h\dot{q}_2$, $\dot{M}_{22} = 0$:

$$\tfrac{1}{2}\dot{q}^\top\dot{M}\dot{q} = \tfrac{1}{2}\big(-2h\dot{q}_2\dot{q}_1^2 - 2h\dot{q}_2\dot{q}_1\dot{q}_2\big) = -h\dot{q}_1^2\dot{q}_2 - h\dot{q}_1\dot{q}_2^2,$$

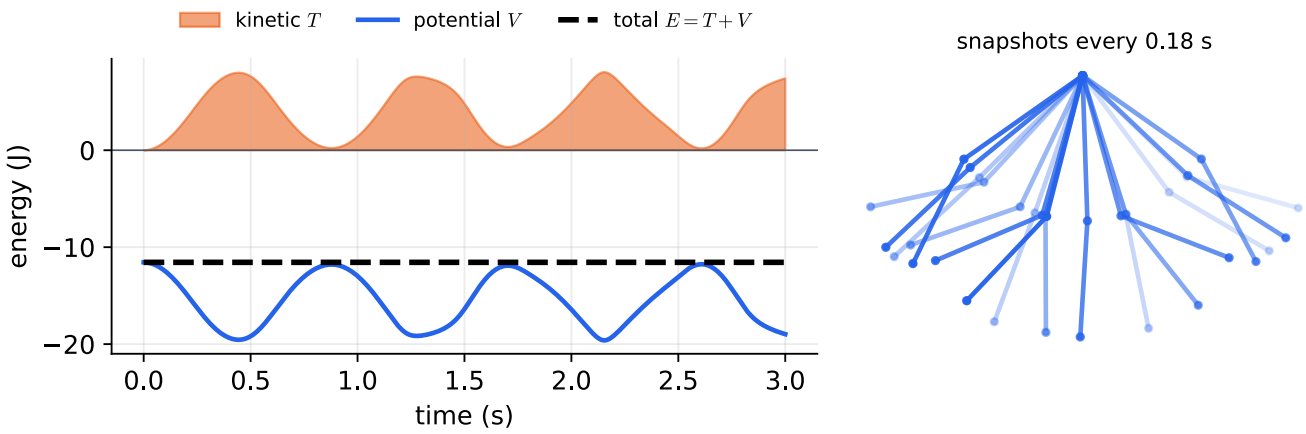
$$\dot{q}^\top c = \dot{q}_1\big(-2h\dot{q}_1\dot{q}_2 - h\dot{q}_2^2\big) + \dot{q}_2\big(h\dot{q}_1^2\big) = -h\dot{q}_1^2\dot{q}_2 - h\dot{q}_1\dot{q}_2^2. \;\checkmark$$

Both are equal, so the bracket vanishes. (In matrix language: $\dot{M} - 2C$ is skew-symmetric, so $\dot{q}^\top(\dot{M} - 2C)\dot{q} = 0$. The Coriolis terms do *no work*: they only move energy between the two joints, never create it.) What remains is

$$\boxed{\frac{dE}{dt} = \underbrace{\dot{q}_1\tau_1 + \dot{q}_2\tau_2}_{\text{power from the motors}} - \underbrace{b\,(\dot{q}_1^2 + \dot{q}_2^2)}_{\text{power burned by friction } \geq 0}}$$

KEY RESULT

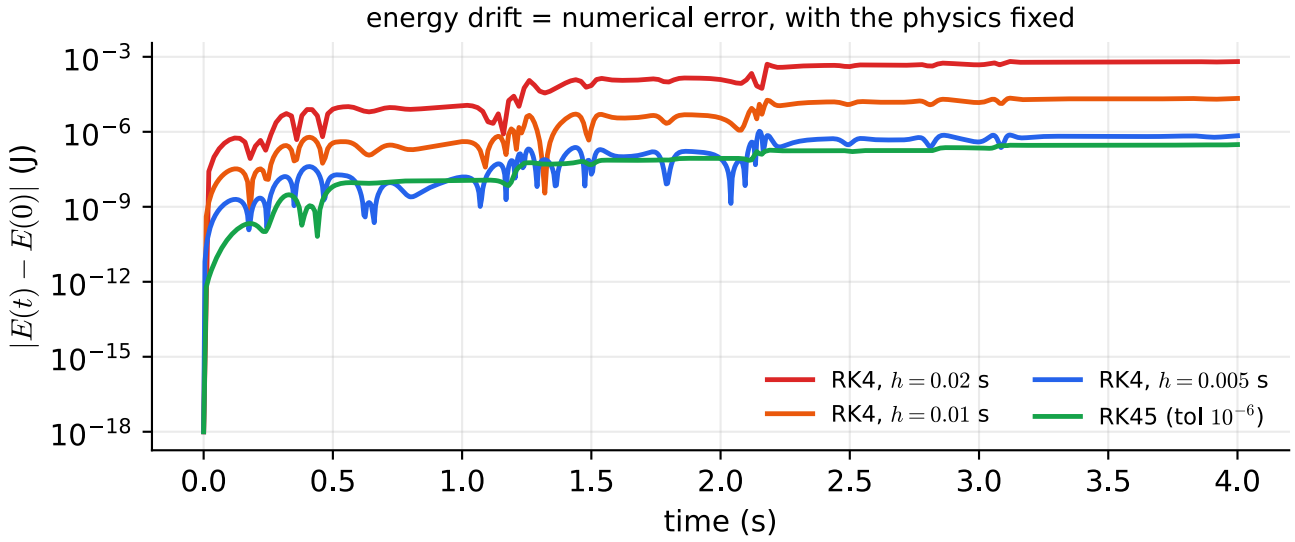
No motor, no friction ($\tau = 0,\ b = 0$) $\Rightarrow dE/dt = 0$: energy is conserved. With friction alone, $dE/dt = -b|\dot{q}|^2 \leq 0$: energy only decreases. With a motor, E goes up when τ and $\dot{q}$ point the same way (the motor pushes the joint along) and down when they oppose. The product "torque $\times$ angular velocity" is the mechanical power, like force $\times$ velocity.



Energy exchange in a swinging, frictionless arm (no motor, $m_p = 0.5$ kg, released from rest). The orange area is the kinetic energy and the blue line is potential: they trade places as the arm swings, but their sum (dashed) stays flat: it varies by only 4.3e-08 J over 3 s. The snapshots on the right show the motion.

4 Energy as a debugging tool

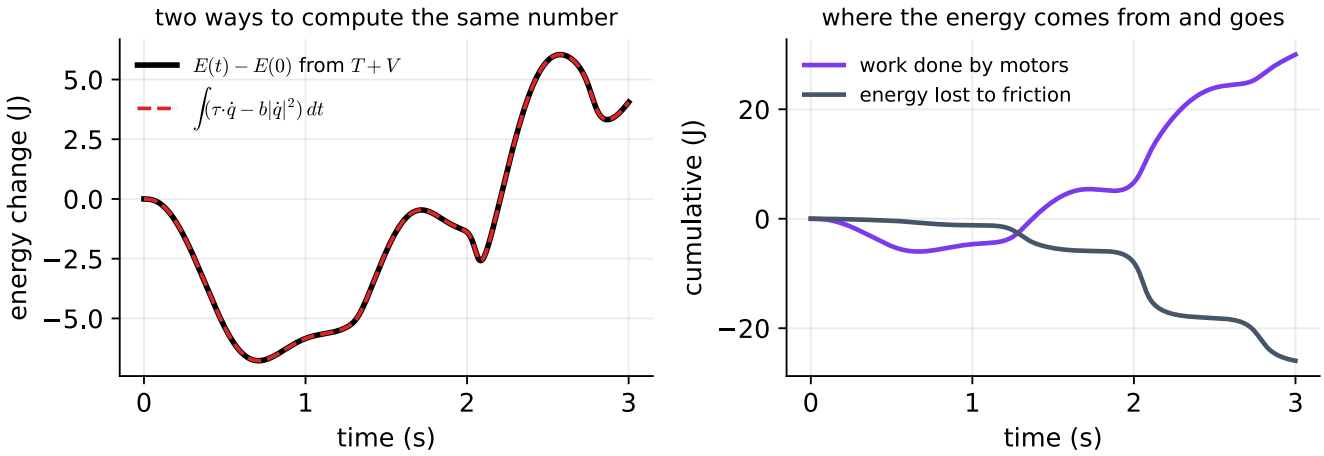
Because the exact solution conserves E when $\tau = b = 0$, any *measured* drift is an error, either in the formulas (a wrong sign in c or G makes energy appear or vanish) or in the integrator. This is exactly what `test_energy_conserved_without_friction_or_torque` checks. A numerical method has a small error even when the model is perfect:



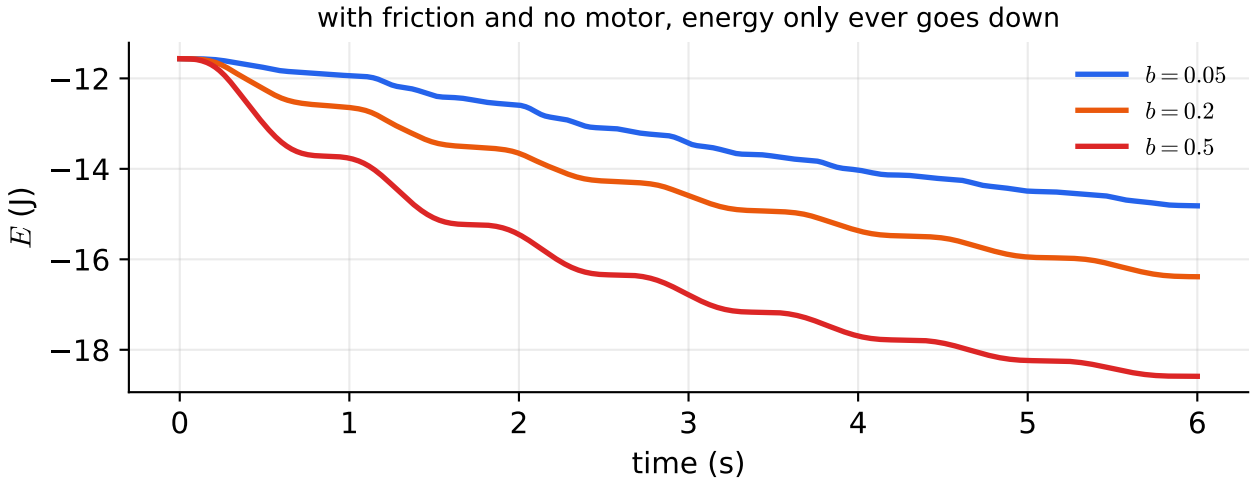
Energy drift of different solvers on the frictionless arm. Each halving of the RK4 step shrinks the drift by a factor of roughly 20 to 30 (ratios about 29 and 21; a 4th-order method promises at least $2^4 = 16$): max drift 6.5e-04, 2.2e-05 and 1.1e-06 J for $h = 0.02, 0.01, 0.005$. So the physics code is right *and* the integrator behaves as advertised.

5 Checking the power balance with friction and a motor

We can also test the *full* law. Simulate with $b = 0.3$ and sinusoidal torques, and compare the actual change of E with the time integral of the right-hand side of the boxed formula:



Energy bookkeeping. Left: the change in E computed from the state (black) is indistinguishable from the integrated power balance (red dashed); largest deviation 3.1e-04 J. Right: the motors pump energy in (purple) while friction drains it (grey).



Friction only. Released from a raised pose with no motor torque the energy decays monotonically, faster for larger b ($dE/dt = -b|\dot{q}|^2$). Each curve is heading towards the value for the arm hanging at rest (a negative V).

6 Worked example with real numbers

$m_p = 0.5$ kg gives $m_{2e} = 2.000$, $l_{c2e} = 0.250$. State $x = [0.5,\ 0.8,\ 2.0,\ -1.0]$.

part	calculation	value (J)
T	$\frac{1}{2}(1.1750)(2)^2 + (0.3342)(2)(-1) + \frac{1}{2}(0.1600)(-1)^2 = 2.3500 + (-0.6684) + 0.0800$	1.7617
V	$9.81\,[\,2.0 \cdot 0.25 \cdot \sin 0.5 + 2.000(0.5 \sin 0.5 + 0.250 \sin 1.3)\,]$ with $\sin 0.5 = 0.4794$, $\sin 1.3 = 0.9636$	11.7810
$E = T + V$		13.5427

This equals `energy(x, p)`.

7 Code ↔ maths dictionary

code	maths
`0.5 * qdot @ mass_matrix(q, p) @ qdot`	$T = \frac{1}{2}\dot{q}^\top M \dot{q}$
`p.m1*p.lc1*np.sin(q1)`	$m_1 g y_1 / g$: height of COM 1
`m2e*(p.l1*np.sin(q1) + lc2e*np.sin(q1+q2))`	$m_{2e}\, y_2 / g$: height of COM 2 (elbow height + rise along link 2)
`p.g * ( ... )`	$V = g \sum m\, y$
`return float(T + V)`	$E = T + V$

CHECK YOURSELF

1. The arm hangs still at $q = (-\pi/2, 0)$. Is E zero? *Answer:* No: $T = 0$ but $V = g[-m_1 l_{c1} - m_{2e}(l_1 + l_{c2e})] < 0$. Zero of V is at joint-1 height, not at the lowest point.
2. A motor applies $\tau_1 > 0$ while joint 1 rotates backwards ($\dot{q}_1 < 0$). Does it add or remove energy? *Answer:* $\tau_1 \dot{q}_1 < 0$: it removes energy (it brakes).
3. Why does the code not use I_{2e} in V? *Answer:* gravitational energy depends only on mass and COM height; inertia only affects motion energy, via M.